

Advanced (Carolina Reaper)

- Q1.** Which of the following is a relation but not a function?
- (A) A set of ordered pairs with unique first elements
 - (B) A set of ordered pairs with repeating first elements
 - (C) A mapping of elements from one set to another
 - (D) A graph that passes the vertical line test
 - (E) A table of values with input-output pairs
- Q2.** The domain of the function $f(x) = \frac{1}{x}$ is
- (A) All real numbers
 - (B) All real numbers except $x = 0$
 - (C) Only positive real numbers
 - (D) Only negative real numbers
 - (E) The set of integers
- Q3.** The range of the function $f(x) = x^2$ is
- (A) The set of all real numbers
 - (B) The set of all non-negative real numbers
 - (C) The set of all non-positive real numbers
 - (D) The set of all integers
 - (E) The set of all rational numbers
- Q4.** The function $f(x) = 2x + 1$ can be written in the form $f(x) =$
- (A) $x \cdot 2 + 1$
 - (B) $x + 2 \cdot 1$
 - (C) $2 \cdot x + 1$
 - (D) $2x - 1$
 - (E) $x^2 + 1$
- Q5.** What is the value of $f(2)$ for the function $f(x) = x^2 - 1$?
- (A) 3
 - (B) 5
 - (C) 2
 - (D) 1
 - (E) 0
- Q6.** The graph of the function $f(x) = |x|$ has a
- (A) Minimum value at $x = 0$
 - (B) Maximum value at $x = 0$
 - (C) Minimum value at $x = 1$
 - (D) Maximum value at $x = 1$
 - (E) Minimum value at $x = -1$
- Q7.** The function $f(x) = \frac{1}{x-1}$ has a
- (A) Horizontal asymptote at $y = 0$
 - (B) Vertical asymptote at $x = 0$
 - (C) Vertical asymptote at $x = 1$
 - (D) Horizontal asymptote at $y = 1$
 - (E) Oblique asymptote at $y = x$
- Q8.** The function $f(x) = 2^x$ is an example of a
- (A) Linear function
 - (B) Quadratic function
 - (C) Exponential function
 - (D) Polynomial function
 - (E) Rational function

Open Ended Questions (FRQ)

- Q9.** Find the domain and range of the function $f(x) = \sqrt{x-1}$ and explain your reasoning.
- Q10.** Determine whether the relation $f(x) = |x-2|$ is a function and explain why or why not. Then, find the domain and range of the relation.

Chef's Tips

- Tip 1: Use the vertical line test to determine if a relation is a function.
- Tip 2: Identify the domain and range from the graph or table of a function.
- Tip 3: Use proper function notation when evaluating functions.